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Model Pratical Exam

1.SQL JOIN

Input:

```
SELECT
    p.name AS patient_name,
    d.name AS doctor_name,
    d.specialization,
    a.app_date
FROM Patients p
JOIN Appointments a
    ON p.patient_id = a.patient_id
JOIN Doctors d
    ON a.doctor_id = d.doctor_id
WHERE a.status = 'Completed';
```

Output:

patient_name	doctor_name	specialization	app_date
Jashwanth	Dr. Priya	Cardiologist	2026-08-25
Leeba	Dr. Arun	Dermatologist	2026-08-28
Rahul Kumar	Dr. Priya	Cardiologist	2026-08-30

2. CREATE VIEW

Input:

```
CREATE VIEW vw_unpaid_bills AS

SELECT
    mb.bill_id,
    p.name AS patient_name,
    d.name AS doctor_name,
    mb.bill_amount
FROM MedicalBills mb
JOIN Appointments a
    ON mb.appointment_id = a.appointment_id
JOIN Patients p
    ON a.patient_id = p.patient_id
JOIN Doctors d
    ON a.doctor_id = d.doctor_id
WHERE mb.payment_status = 'Pending';
```

Output:

<

3. User Defined Function

Input:

DELIMITER //

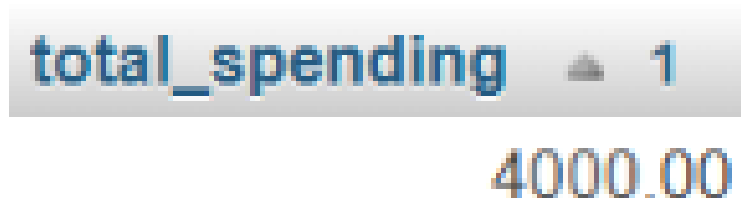
```
CREATE FUNCTION fn_total_patient_spending(p_patient_id INT)
RETURNS DECIMAL(10,2)
DETERMINISTIC
READS SQL DATA
BEGIN
    DECLARE total_spending DECIMAL(10,2);

    SELECT COALESCE(SUM(mb.bill_amount), 0)
    INTO total_spending
    FROM MedicalBills mb
    JOIN Appointments a
        ON mb.appointment_id = a.appointment_id
    WHERE a.patient_id = p_patient_id
        AND mb.payment_status = 'Settled';

    RETURN total_spending;
END //
```

DELIMITER ;

Output:



total_spending	1
4000.00	

4. Trigger

Input:

```
DELIMITER //

CREATE TRIGGER trg_prevent_doctor_deletion

BEFORE DELETE ON Doctors

FOR EACH ROW

BEGIN

    IF EXISTS (

        SELECT 1

        FROM Appointments

        WHERE doctor_id = OLD.doctor_id

            AND status <> 'Completed'

            AND app_date >= CURDATE()

    ) THEN

        SIGNAL SQLSTATE '45000'

        SET MESSAGE_TEXT =

            'Cannot delete doctor: active uncompleted appointments exist.';

    END IF;

END //

DELIMITER ;
```

Output:

Error

SQL query: [Copy](#)

```
DELETE FROM Doctors
WHERE doctor_id = 283;
```

MySQL said: 

#1644 - Cannot delete doctor: active uncompleted appointments exist.